

Lorenzo Valente

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EDUCATION

By March 2027	PhD in Physics University of Hamburg, Germany Research Focus: Machine Learning for Fast Calorimeter Simulation, in collaboration with DESY. Supervisor: Prof. Gregor Kasieczka, co-supervisor Dr. Frank Gaede
March 2023	Master in Nuclear and Subnuclear Physics University of Bologna, Italy Thesis title: “A variational autoencoder application for real-time anomaly detection at CMS”. Supervisor: Prof. Daniele Bonacorsi
July 2020	Bachelor in Physics University of Bologna, Italy Thesis title: “Reconstruction of non-prompt charmed baryon Λ_c with Boosted Decision Trees technique” (ALICE experiment). Supervisor: Prof. Andrea Alici
October 2018	Master in Classical Piano Conservatory of Music Luisa D’Annunzio, Pescara, Italy

PUBLICATIONS

* *corresponding author*

- F. Gaede, G. Kasieczka, **L. Valente***, *Cross-Geometry Transfer Learning in Fast Electromagnetic Shower Simulation*, JINST **21** (2026) P07037.
- H. Day-Hall*, ..., **L. Valente**, *CaloClouds3: Ultra-fast Geometry-Independent Highly-Granular Calorimeter Simulation*, JINST **21** (2026) P03018.
- T. Buss, ..., **L. Valente***, *Transferable Fast Calorimeter Shower Generation via Multi-Geometry Pre-training*, arXiv:2608.18233 (2026), alphabetical authors, led study design, dataset production, training, and analysis.
- **L. Valente*** et al., *Cross-Geometry Fast Electromagnetic Shower Simulation*, proceedings of ACAT 2025, submitted to PoS.
- **L. Valente*** et al., *Joint Variational Auto-Encoder for Anomaly Detection in High Energy Physics*, PoS ISGC&HEPiX2023 (2023) 014.

TALKS

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| ○ European AI for Fundamental Physics Conference (EuCAIFCon) | Heidelberg, Germany |
| “Cross-Geometry Shower Simulation via Pre-training and Transfer” | 24 August 2026 |
| ○ 14th Large Hadron Collider Physics Conference (LHCP) | Paris, France |
| “ML and Detector Simulation” (invited) | 21 May 2026 |
| ○ 23rd International Workshop on Advanced Computing and Analysis Techniques in Physics Research (ACAT) | Hamburg, Germany |
| “Cross-Geometry Fast Electromagnetic Shower Simulation” | 11 September 2025 |
| ○ 7th Inter-Experimental LHC Machine Learning Workshop | CERN, Switzerland |
| “Cross-Geometry Fast Electromagnetic Shower Simulation” | 20 May 2025 |

○ **DPG Spring Meeting 2025**

“Adaptive Generative Modeling for Accelerated Calorimeter
Simulations via Domain Transfer”

Göttingen, Germany

3 April 2025

RESEARCH EXPERIENCE

- October 2023 – present **Doctoral Researcher**
University of Hamburg, Germany
Machine learning for fast calorimeter simulation, based on generative models
and on transfer learning to unseen detector geometries.
Supervisors: Prof. Gregor Kasieczka and Dr. Frank Gaede, in collaboration
with Dr. Katja Krüger
- April – July 2023 **Summer Student**
Instituto de Astrofísica de Canarias (IAC), Spain
Deep learning classification of heavy nuclei in cosmic rays (iron spectrum),
ensemble approach reaching AUC 0.90.
Funded by an Erasmus+ traineeship grant.
Supervisors: Dr. Mónica Vázquez Acosta and Dr. Miguel Molero González
- June 2022 – March 2023 **Master’s Thesis Research**
University of Bologna, Italy
Variational autoencoder for real-time anomaly detection at the CMS Level
1 trigger on public simulation, quantisation-aware training with QKeras and
hls4ml.
Supervisor: Prof. Daniele Bonacorsi

TEACHING

- AI in Physics**, MSc course, University of Hamburg, tutor, 2025
- Python tutorial**, BSc course, University of Hamburg, tutor, 2025
- BSc thesis co-supervision**, Investigating the Behavior of Generative Models in Particle Physics
under Recursive Training, University of Hamburg, 2025

SCHOOLS AND WORKSHOPS

- ELLIS Winter School on Foundation Models**, Amsterdam, Netherlands March 2025
- Generative Modeling Summer School**, Eindhoven, Netherlands July 2024
- Machine Learning with FPGAs for Particle Physics**, Bologna, Italy November 2022
- Quantum Sensing, Information Processing and Computing**, Bologna, Italy July 2022

TECHNICAL SKILLS

- Simulation & HEP Software:** ROOT, Geant4, DD4hep (ddsim, k4geo, ddfastsim), Key4hep,
LCIO, uproot, awkward
- Programming Languages:** Python, C/C++, Bash, LabVIEW
- Machine Learning:** PyTorch, TensorFlow, Keras, scikit-learn, ONNX export, mixed-precision
training, experiment tracking (Comet)
- Computing Infrastructure:** Slurm (DESY Maxwell), HTCondor (DESY NAF), Singularity con-
tainers, distributed multi-GPU training, HDF5
- Real-Time & FPGA:** hls4ml, QKeras, Vivado HLS, VHDL
- Software Engineering:** Git, CI, pytest, Docker

LANGUAGES

ITALIAN (Native), ENGLISH (Fluent), SPANISH (B2), GERMAN (B1)